When Memory Metrics Hide Memory Policies

Same-Stream Evaluation and Policy-Facing Lookup Contracts for Persistent-Agent Memory

Manav Malhotra

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Abstract

Agent-memory benchmarks often ask whether a system produced the right answer or retrieved relevant text. This paper asks a narrower policy question: whether the memory system exposes the governable evidence handle needed to contest, scope, repair, and audit the state it uses. We present a preregistered attribution discipline for persistent-agent memory: hold the upstream candidate stream and scoped storage substrate fixed, gate noisy comparisons on extractor quality, and add policy-facing lookup-contract (PFLC) diagnostics for handle identity.

The method exposes hidden structure at two layers. At the extractor layer, an aggregate-passing 7B component row would have falsely authorized downstream policy comparison, while the full gate blocks it and licenses only the 32B cells. At the memory-state layer, PFLC shows why clustering, retrieval, and answer metrics can miss whether the policy-facing handle itself is available. A memory can contain the right text but still fail if the policy cannot name the exact stored claim it needs to retract, scope, or repair. A published-output replay on AMB LoCoMo decomposes aggregate answer accuracy into evidence exposure, rank/context saturation, and answer residuals; that replay demonstrates evidence-id exposure, while handle-identity failure is shown by the internal canonical-id diagnostic.

We demonstrate the discipline on Consolidation Queue (CQ), a staged-write policy that keeps low-confidence claims pending and contestable. The case study finds one clean noisy mechanism-survival row, a stable LongMemEval v1 evidence-completeness loss, and a preregistered readout-cardinality repair corroborated by a capped Reflection control. These results support the attribution discipline, not broad CQ superiority or deployment readiness.

1 Introduction

An agent helping a user across months learns small facts incrementally. In March, the user mentions they prefer espresso; in August, they are cutting caffeine and want tea in the morning. A naive memory system overwrites the older preference; a less naive one stores both as parallel facts. Neither resolves the conflict when the user later asks what to order. The policy question for persistent memory is whether the system can keep both claims, scope them, contest one with the other, and demote a durable claim when the evidence shifts.

Most agent-memory benchmarks ask whether the system got the right answer; this report instead asks whether the memory policy had the right governable evidence handle: the stored-claim identifier a policy uses to contest, scope, demote, or repair memory. Recall, retrieval, and answer metrics are important, but they can hide two different failures. Upstream, aggregate extractor scores can authorize a candidate stream—the shared extracted-fact list every policy receives—that collapses the policy comparison. Downstream, memory-state metrics can expose relevant content while failing to certify the policy-facing handle identity needed for contradiction, scope, demotion, and repair. Recent memory benchmarks broaden what systems are tested on; this report narrows how policy-level credit is assigned.

The broader benchmark landscape makes this narrower question timely. STATE-Bench evaluates memory in stateful enterprise workflows, EvoMemBench frames memory as self-evolving state where no single memory form is uniformly sufficient, and MemoryAgentBench includes FactConsolidation as an explicit revision and consolidation target [9, 18, 7]. This report therefore argues for a stricter attribution posture. Memory-policy comparisons should hold the upstream candidate stream fixed and use the same storage substrate, the shared memory store for compared policies. They should separate oracle-mode policy claims (known scope, conflicts, and sources) from noisy-pipeline claims (real extracted inputs). Noisy comparison runs only after an extractor-quality gate, a check that the fact extractor is adequate. Artifacts stay inspectable so failure cases can be audited. The running case study is a lightweight consolidation queue policy (CQ),[1] compared with immediate-write Reflection, an ADD/UPDATE/NOOP eager-write baseline (`ADD/UPDATE/NOOP EagerWrite`, not the production Mem0 system), and ablated variants. We use CQ to test which memory-policy distinctions survive the interfaces around them.

Concretely, CQ is a staged-write policy. Incoming candidate memories enter a queue with scope, provenance, support, contradiction, and lifecycle metadata. High-scoring candidates promote to durable memory; lower-scoring but usable candidates can remain pending and still answer scoped lookups. Contradictory evidence contests earlier candidates and can demote active durable memories, while narrower pending overrides can shadow a wider durable claim without globally overwriting it. In the opening example, CQ keeps both the espresso preference and the morning-tea preference as scoped claims: the narrower morning-tea claim shadows the broader espresso claim at read time, and either can be contested or demoted when stronger evidence arrives.

Internally, under a preregistered noisy policy comparison, where predictions and gates were locked before running, CQ's contradiction-edge-driven contestation and demotion mechanism remains visible on forced contradiction and partially visible on preference drift. Other rows stay recorded as nulls or policy-facing lookup-contract failures, where the policy cannot query the right stored claim. The same discipline also surfaces an extractor false-unlock, where aggregate scores clear but policy comparison should not be licensed. Externally, a controlled LongMemEval v1 readout-cardinality probe, which tests how many evidence sessions are returned, produces a stable negative result for base CQ on evidence completeness. A later preregistered follow-up repairs that specific failure by exposing all eligible same-slot pending evidence at answer time, while a cardinality-capped Reflection control reproduces the original loss. The paper preserves the failure and the repair as separate evidence.

[1] A non-archival design sketch is available at The Consolidation Queue: How Persistent AI Earns Durable Change.

Table 1: Contribution map.

Layer	Contribution
Problem	Memory scores can hide whether policy-level governance succeeded.
Stream control	Same-stream, same-substrate comparison (same inputs, same store) makes policy attribution inspectable.
Extractor gate	Layered component checks block an aggregate-passing false unlock before downstream policy comparison.
Memory-state diagnostic	Identity-PFLC (policy-facing lookup contract; right handle exposed) checks whether the policy-facing handle is exposed, not just whether text is relevant.
Published-output diagnostic	AMB LoCoMo decomposes answer accuracy into evidence exposure, rank/context saturation, and answer residuals (returned gold, context rank, leftover errors).
Case study	CQ shows one clean noisy mechanism-survival row (mechanism stays visible), bounded nulls (inconclusive, not ties), and a preserved LongMemEval v1 evidence-completeness loss.
Repair pattern	A plural-readout repair plus capped Reflection control isolates answer-time readout cardinality (how many items returned).

The contribution is the attribution discipline; CQ is the case study used to test what the discipline can detect. This is an evaluation-methods report, not a benchmark leaderboard or a deployable memory-system release. The paper collapses the repository's detailed evidence ledger into four claims, with the following reader contract.

Table 2: Reader contract for the methods-first reading of this report.

This paper does not claim	Boundary used in the paper
CQ is broadly superior to immediate-write memory systems.	CQ is a case-study policy. The supported noisy result is mechanism-local (limited to named mechanisms): forced contradiction survives cleanly and preference drift survives partially.
The LongMemEval follow-up proves broad external generalization.	The follow-up repairs a v1 evidence-completeness loss caused by answer-time readout cardinality; distractor-heavy retrieval and natural-language QA remain out of scope.
The ADD/UPDATE/NOOP baseline is a full reproduction of production Mem0.	The baseline is an ADD/UPDATE/NOOP published-family policy over the shared candidate stream and substrate, not the production system.
PFLC is a replacement leaderboard metric or a new pass/fail gate.	PFLC is a diagnostic for whether the policy-facing evidence handle is exposed; it supplements clustering, retrieval, and answer metrics. LoCoMo demonstrates evidence exposure over gold evidence ids; handle-identity failure is shown by the internal canonical-id diagnostic (exact stored-claim lookup).

Claim 1: Same-stream evaluation makes policy attribution inspectable. Same-stream, same-substrate, extractor-gated evaluation with preserved lifecycle traces removes the main stream, substrate, and extractor-quality confounds. Noisy policy comparison unlocks only after the extractor-quality gate records sufficient component quality. The benchmark fixes the upstream

`CandidateUpdate` stream and the scoped memory substrate across policies, then records lifecycle traces, manifests, failure examples, and replay checks.

Claim 2: Standard memory metrics can pass while policy-facing lookup fails. Cluster-partition, retrieval-at-k, and answer-accuracy metrics can all be passed by systems that fail exact policy-facing lookup. PFLC asks whether the system exposes the evidence handle the policy must actually query, not just whether retrieved text is semantically relevant. The identity-PFLC gap is the memory-state counterpart of the extractor-gating gap: the aggregate score looks good, but the governance-relevant handle is missing. The field survey shows when released benchmark artifacts do or do not expose the handles needed to replay such diagnostics.

Claim 3: A preserved negative result exposes interface failures. A controlled externalization pipeline that preserves negative results instead of rescuing them can surface stable interface failures that answer-only, any-hit, or semantic-relevance scoring would hide. The LongMemEval workstream builds a redacted same-stream adapter, hidden-answer protocol, judge gate, PFLC scorer, and contract-sensitivity audit before policy execution. The result is stable and negative for base CQ on `all_hit_at_50`, which asks whether all required evidence sessions are returned; the missing second evidence session drives the loss.

Claim 4: A paired repair/control isolates readout cardinality. When a controlled-probe loss is paired with both a single-variable interface repair and a capped-baseline control that reproduces the loss on a winning baseline, the loss can be attributed to an interface — here, answer-time readout cardinality, the number of evidence items returned — rather than to storage asymmetry or input asymmetry. The separately preregistered follow-up exposes plural pending evidence for CQ and caps a winning Reflection baseline to one returned candidate. Both predictions hold, supporting a mechanism-local repair pattern rather than broad external generalization.

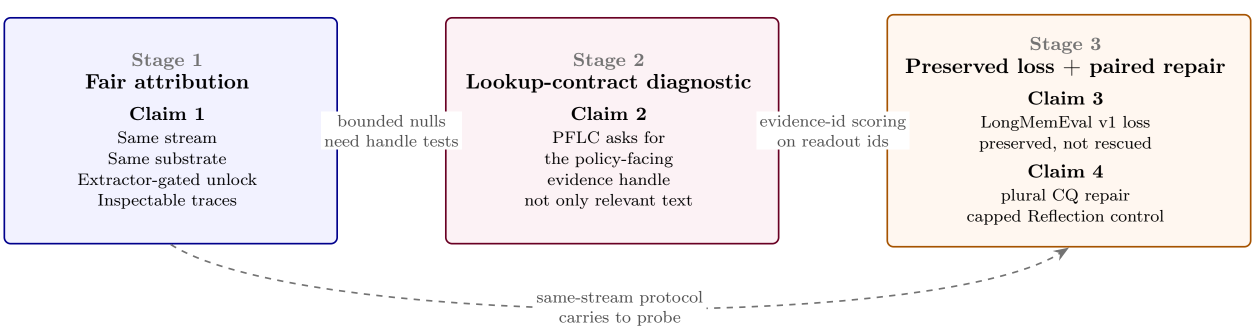

Figure 1: Method-stage view of the four claims. Stage 1 makes policy attribution inspectable by fixing the stream, substrate, and extractor gate. Stage 2 adds PFLC to test whether the policy-facing evidence handle is exposed. Stage 3 preserves the LongMemEval v1 loss and then uses a paired repair/control to isolate answer-time readout cardinality. The figure preserves the claim numbering but frames the contribution as an attribution method.

All empirical claims in this paper cite committed artifacts. Where a result is diagnostic-only, descriptive-only, or blocked by missing artifacts, the paper keeps that boundary in the main text rather than relegating it to a caveat.

2 Related Work

External memory benchmarks. LongMemEval evaluates long-term chat memory across information extraction, multi-session reasoning, knowledge updates, temporal reasoning, and abstention [19]. Here, the released v1 oracle split, a set of cases with known gold evidence structure, becomes a controlled-pilot anchor only after a preregistered adapter creates a redacted same candidate stream for all local policies. The resulting PFLC target is dialog-evidence-id completeness over the v1 evidence-only oracle split, so the result should be read as evidence exposure (whether returned items include the gold evidence ids) and completeness under a fixed policy surface, with distractor-heavy retrieval quality over LongMemEval-S or LongMemEval-V2 left to future work. LoCoMo is the other key long-conversation benchmark used in the PFLC survey: it collects very long multimodal conversations and evaluates question answering, event summarization, and multimodal dialogue generation [10].

MemoryAgentBench and MemBench are important external-evaluation references because they stress incremental interaction, long-range memory, factual memory, and selective forgetting [7, 17]. MemoryAgentBench's FactConsolidation task makes revision and consolidation an explicit benchmark target. These benchmarks occupy the benchmark lane, separate from this report's fair-policy-comparison lane, because standard benchmark outputs need not enforce same-candidate-stream and same-substrate constraints. The PFLC survey treats their released artifacts as a diagnostic question: do standard outputs expose the per-question identifiers needed to test whether a memory policy can query the right handle?

Recent and concurrent benchmark efforts broaden this lane further. STATE-Bench measures whether memory improves reliability, task completion, efficiency, and user experience in stateful enterprise tasks [9]. MemGym adds long-horizon agentic tracks across tool-use dialogue, deep research, coding, and computer-use settings [22]. EvoMemBench evaluates self-evolving memory across in-episode/cross-episode and knowledge/execution-oriented axes [18]. MemoryArena couples memory with later action in interdependent multi-session agent-environment loops, rather than testing recall alone [6]. AMA-Bench moves from dialogue-centric memory to long-horizon trajectories of states, actions, observations, and tool outputs [23]. Scale-conditioned evaluation holds task evidence fixed while adding irrelevant sessions, making claims conditional on scale, agent, interface, and interaction budget [15]. The broader survey literature similarly describes agent memory as a write–manage–read loop coupled to perception and action [5]. These benchmarks and surveys broaden what memory systems are tested on; this report narrows how policy-level credit is assigned.

Table 3: Benchmark surfaces and this paper's diagnostic layer.

Surface	What it emphasizes	Boundary in this paper
STATE-Bench / Mem-Gym / EvoMemBench / MemoryArena / AMA-Bench	Stateful, long-horizon, self-evolving, and action-coupled memory tasks	Context for why memory evaluation is broadening; not evidence for CQ.
Scale-conditioned evaluation	Fixed evidence with added irrelevant sessions	Adjacent evidence-control framing; this paper instead fixes policy inputs and storage to attribute policy-level memory governance.
MemoryAgentBench FactConsolidation	Revision and consolidation as explicit benchmark targets	Future transfer target; no same-stream policy run here.
AMB LoCoMo replayable outputs	Per-question context with recoverable gold evidence ids	Evidence-exposure decomposition, not handle-identity certification.
This work	Same-stream policy isolation plus extractor-gated noisy comparison	One clean mechanism-survival row, a blocked false unlock, LoCoMo evidence exposure, and an internal canonical-id handle-identity diagnostic.

The answer splits. Most surveyed released outputs are artifact-blocked, meaning public outputs lack per-question ids for replay, but published Agent Memory Benchmark (AMB) LoCoMo runs (Hindsight, hybrid-search, Cognee) inject context with recoverable LoCoMo `dia_ids` and admit a published-output PFLC replay, which decomposes aggregate answer accuracy into evidence exposure, rank, and answer-generation residuals (Section 4.4).

Belief revision and truth maintenance. CQ's contestation and demotion behavior is adjacent to classical truth maintenance and belief-revision work, which tracked reasons for beliefs, assumption sets, and revision under inconsistent or changing evidence [4, 3, 1]. This paper does not claim a new general belief-revision logic. It uses that lineage as a local memory-policy pattern for LLM agents, adding scoped candidate streams, queryable pending state, and same-stream comparison against eager-write policies.

Memory systems and write policies. Mem0 proposes a production-oriented memory architecture with dynamic memory extraction, update, and retrieval, including a graph variant for relational memory [2]; its current documentation also exposes add, search, update, and delete operations [12]. A-MEM emphasizes dynamic indexing, memory linking, and memory evolution [21]. MemGPT/Letta frames stateful agents through virtual context management and self-editing memory tiers [13]. Zep/Graphiti and Cognee represent graph-oriented memory and retrieval stacks that motivate the related-system lane without defining this report's fair policy-comparison surface [14, 11]. These systems motivate comparison targets and terminology. The local `ADD/UPDATE/NOOP EagerWrite` baseline is a narrow ADD/UPDATE/NOOP write-time policy over the same candidate stream and substrate as CQ and `ReflectionEagerWriteLite`; that narrowness keeps the policy comparison inspectable. `ADD/UPDATE/NOOP EagerWrite` is *not* an end-to-end reproduction of the Mem0 system, and our results do not characterize production Mem0.

Lookup-style memory boundaries. Recent positioning work argues that many agentic memory systems behave more like memo or lookup substrates than true long-term learning systems [20]. This report uses that claim only as boundary-setting context. Even memo-like stores need auditable identifiers, provenance, contradiction handling, and repair surfaces; otherwise errors can propagate downstream through future retrieval and policy decisions.

Memory governance layers. Recent governance-oriented proposals also identify contradiction, stale memory, semantic drift, and privacy as memory-management risks. SSGM frames governed memory as a safety architecture with verification, temporal decay, and dynamic access control before consolidation, while MemArchitect proposes a policy layer for decay, conflict resolution, and privacy controls [8, 16]. This report is complementary: it does not propose a general deployment governance framework, but tests how to attribute policy-level credit when those governance mechanisms are evaluated. The same-stream, same-substrate, extractor-gated protocol and PFLC diagnostics are therefore evaluation controls for governed memory, not a competing governance architecture.

Preregistration and audit discipline. The methodological stance is closer to a preregistered evaluation audit than to a recall leaderboard. The project locks prediction grids, outcome buckets, adapter pins, judge gates, and follow-up boundaries before running each major comparison. It also records aborts as evidence. The canonical-id resolution audit, for example, refused to emit a verdict twice when locked run JSON payloads could not be reproduced. That posture prevents a methodology repair from becoming a post-hoc result rescue.

This is not a claim that preregistration itself is novel to machine learning. The ML Reproducibility Challenge, preregistration tracks hosted through OpenReview venues, and MLPerf-style standardized evaluation streams are all part of the existing methodological toolkit. The contribution here is narrower: PFLC as a metric-class concept for memory governance, same-stream and same-substrate controls applied to persistent-memory policy comparisons, and a case study that preserves a negative externalization result while using paired controls to isolate a specific interface mechanism.

3 Method

3.1 Notation and Evaluation Stages

The core unit is a `CandidateUpdate`: a proposed memory item with claim content, scope key, provenance, evidence links, and lifecycle state. A *durable* memory is committed state. A *pending* candidate is not durable, but CQ can still expose it to scoped lookup while it remains reversible. A *same-stream* comparison gives every policy the same ordered `CandidateUpdate` stream. A *same-substrate* comparison stores every policy's state through the same scoped memory interface.

PFLC means policy-facing lookup contract: the identifier or handle a memory policy must expose so a query can be evaluated against the intended evidence. The *canonical-id lookup diagnostic* is this paper's canonical-id PFLC instance. It reuses the set-membership metric and alias function from the earlier *canonical-id resolution audit*, but it is a small committed diagnostic table rather than a replay of the large noisy same-stream policy comparison run payloads.

This paper uses descriptive names in body text. Appendix A collects the repository-label mapping for artifact navigation. The evaluation has four steps:

- The *prompt-regression guard* runs before noisy component scoring.

- The *noisy extractor-quality gate* scores local extractor outputs and licenses downstream policy comparison.
- The *noisy same-stream policy comparison* runs after that gate passes.
- The *cardinality-repair follow-up* addresses a specific LongMemEval v1 interface gap.

Preregistered interpretation labels distinguish a planned gate or thesis condition firing, a completed mixed or partial result, where only some mechanism rows survive, and an abort or surprise-lane boundary that prevents the stronger claim. The 7B and 32B labels denote local extractor model cells, not memory-policy sizes.

3.2 Benchmark Objects

The benchmark centers on scoped candidate updates and a shared memory substrate. Candidate updates carry claim content, scope, provenance, evidence relations, and lifecycle state. All policies receive the same upstream observations. They differ in when they promote, demote, contest, or expose these candidates. The load-bearing invariant is that CQ and the immediate-write baselines compare memory policy over the same inputs.

The principal policies are `ConsolidationQueueLite`, `ReflectionEagerWriteLite`, the ADD/UPDATE/NOOP eager-write baseline (`ADD/UPDATE/NOOP EagerWrite`), `NaiveEagerWriteLite`, `NoMemoryLite`, and CQ ablations that remove contestation/demotion, pending lookup, source-independence gating, or wider-scope pending override. Two follow-up policies are explicitly outside the original headline set: an oracle-mode dated-evidence repair and the LongMemEval plural-pending readout repair. The cardinality-repair follow-up also includes a Reflection readout-cap control as a diagnostic control. Appendix A maps these reader-facing names to repository labels.

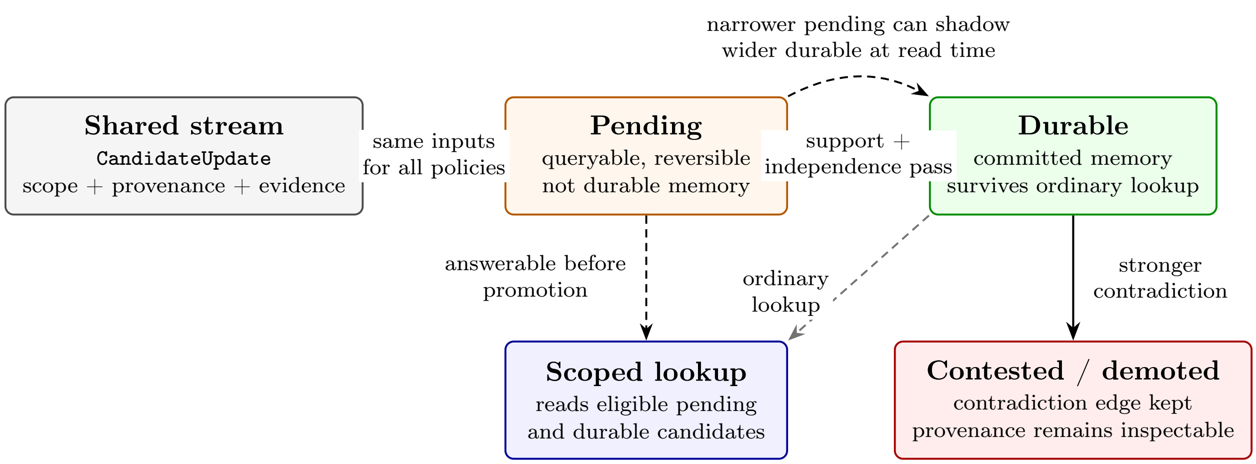

Figure 2: CQ lifecycle. Candidate updates come from the same upstream stream used by every policy in the comparison. CQ differs from eager writing by allowing a candidate to stay pending while still being queryable, by promoting only when support is strong enough, and by preserving contradiction edges so an active durable memory can be contested or demoted. A narrower pending candidate may shadow a wider durable claim for scoped lookup without globally overwriting it.

CQ trace example. Suppose a candidate says the user prefers espresso under the coffee-preference scope. With one source and moderate support, CQ can keep the candidate pending: it is answerable to scoped coffee queries, but it has not become durable memory. Later, a narrower morning-coffee candidate says the user prefers tea in the morning and carries stronger support plus a contradiction edge. CQ records the conflict, lets the narrower pending candidate shadow the broader coffee claim for morning lookups, and, if an older espresso claim had already promoted, demotes that durable record into contested state instead of deleting it. The record's source history remains inspectable. The eager baseline receives the same candidate stream, but writes immediately.

Offline replay and deployment timing. The staged-write policy is not a latency claim. In a deployed agent, extraction and provisional insertion may run on the request path, while support counting, contradiction checks, demotion, and promotion may run asynchronously. The contract tested here is read-time consistency: every policy sees the same candidate stream, lifecycle state is explicit, and eligible pending candidates are exposed according to the policy's read interface. The experiments are offline replay and do not measure production latency.

3.3 Oracle Stage

The oracle stage tests policy behavior when candidate observations, scope keys, contradiction/support relations, and source metadata are known. It includes forced contradiction, scope contamination, preference drift, useful pending memory, false corroboration, memory poisoning, mechanism-diverse frozen held-out scenarios, adversarial upstream noise, and the evidence-conflict spectrum. This stage establishes policy behavior under controlled evidence before extraction quality enters the experiment.

3.4 Noisy Stage

The noisy stage renders transcripts, scores local extractor outputs against component labels, converts accepted predictions into shared candidate streams, and only then runs policies. The component gate has layered checks: aggregate confidence-bound gates, per-family observed gates, frozen mechanism-diverse sentinels, held-out checks that must not regress, prompt-regression checks, deterministic replay, and exact adapter pin validation. The 7B anchor passed aggregate gates but stayed locked because it had concentrated scenario errors and family/sentinel failures. The preregistered 32B cells cleared the gate and licensed the separate noisy same-stream policy comparison.

3.5 Attribution Stage

Policy results require separate attribution. The mixed-support mechanism audit reads policy-comparison artifacts, component-eval artifacts, ablation patterns, exact policy-query canonical-id alignment, and representative traces. It attributes forced contradiction cleanly to contradiction-edge preservation and contestation/demotion. Preference drift is partial survival: the policy delta remains, but absolute CQ performance drops relative to oracle and the 32B artifacts show residual drift. Other rows remain null, descriptive, or policy-facing lookup-contract failures unless an artifact directly supports a stronger claim.

3.6 Policy-Facing Lookup Contracts

PFLC is operational rather than a metric nicety because the evaluated governance operations require identifier-bearing state, not just relevant content. The operations in this paper are contradiction

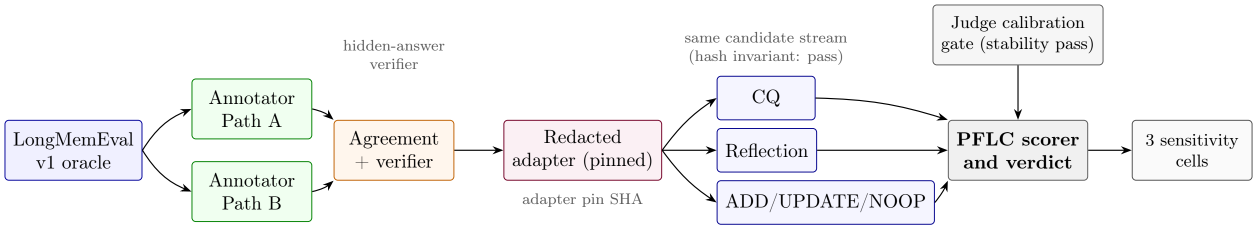

Figure 3: Controlled LongMemEval externalization. The v1 oracle source is annotated by two independent paths whose agreement is verified before any policy sees data. The redacted adapter strips the gold answer, pins a SHA-locked candidate stream, and feeds the same stream to every policy. The judge calibration gate must clear stability before scoring runs; the PFLC scorer reads policy outputs across three contract-sensitivity cells.

handling, contestation, demotion, provenance preservation, and scoped override: each must address a specific prior claim by handle, not by paraphrase or similarity.

PFLC diagnostics ask whether the system exposes the identifier or handle that a memory policy actually needs to query. This identity-PFLC gap is why the title says memory metrics can hide memory policies: a clustering metric treats labels as interchangeable, so a system that perfectly groups items into clusters $\{A, B, C\}$ scores 1.00 even if it called them $\{1, 2, 3\}$ internally. A governance query that asks "is item X in slot B?" fails when the system has only slot 2. Retrieval-at-k and answer-accuracy metrics have analogous gaps: they can expose useful content or a correct answer without certifying the policy-facing handle. Appendix B proves the cluster-partition gap formally; the retrieval and answer gaps are analogous constructions, and the body uses the diagnostic as a named memory-state check.

The lookup diagnostic instantiates this PFLC class on canonical-id queries within the internal audit. It reuses the locked resolution audit's Section A set-membership metric and alias function, reads a committed per-question source table, and reports Wilson intervals (standard binomial confidence intervals) plus a joint B-cubed/lookup table (clustering agreement paired with exact lookup). It is diagnostic and leaves the noisy same-stream policy comparison verdict unchanged.

3.7 Controlled Externalization

The LongMemEval workstream adapts an external benchmark into a controlled same-stream pilot. The protocol selects LongMemEval v1 as the primary anchor, freezes dual-path annotations, verifies the hidden-answer boundary, pins the adapter, calibrates a local judge, and uses PFLC scoring over answer-session ids. The v1 oracle split is evidence-only, so the PFLC metric measures whether policies preserve and expose gold evidence sessions they received, not whether they search through distractor haystacks.

The LongMemEval v1 readout-cardinality probe uses `all_hit_at_50` as the headline metric and preserves `answer_correct` as diagnostic-only because policy answers are structured memory traces. The follow-up keeps the same adapter, judge, and headline metric but changes the policy-facing readout interface under a new preregistration: CQ exposes all eligible same-slot pending candidates, and capped Reflection returns only one durable candidate id at answer time.

4 Results

The claim-level ledger below maps the paper's four claims to their principal evidence anchors. The full artifact index is in Appendix F; this section emphasizes each result's supported scope.

Claim-level evidence ledger.

1. **Same-stream evaluation makes policy attribution inspectable.** Evidence: frozen oracle sweep, noisy extractor-quality gate, and noisy same-stream policy comparison. Readout: all 108 frozen oracle deltas matched, the 7B gate blocked a false unlock, and the 32B-gated comparison reaches mixed/partial support. Boundary: mechanism-local noisy support.
2. **Standard memory metrics can pass while policy-facing lookup fails.** Evidence: internal handle-identity diagnostic, PFLC field survey, and AMB LoCoMo published-output replay. Readout: five of seven comparison rows pass the B-cubed floor while exact canonical-id match is below 5 percent; LoCoMo decomposes evidence-id exposure, rank, and answer residuals. Boundary: identity-PFLC is internal; LoCoMo is evidence exposure.
3. **Preserved negative externalization exposes interface failures.** Evidence: LongMemEval v1 readout-cardinality probe. Readout: methodology gates pass, but CQ minus Reflection on `all_hit_at_50` is -1.0 in all three cells. Boundary: stable base-CQ evidence-completeness loss.
4. **A paired repair/control isolates readout cardinality.** Evidence: pending multi-evidence follow-up. Readout: plural pending readout moves base CQ from 0/71 to 71/71 on the primary cell; capped Reflection falls from 71/71 to 0/71. Boundary: interface repair; external generalization is out of scope.

Two empirical anchors carry the methods-first reading. First, the extractor gate blocks an aggregate-passing 7B row that would have falsely licensed downstream policy comparison. Second, the AMB LoCoMo replay decomposes published answer accuracy into evidence-id exposure, rank/context saturation, and answer residuals. Identity-PFLC remains the memory-state diagnostic: the LoCoMo replay tests evidence exposure over gold evidence ids, while handle identity is demonstrated by the internal canonical-id diagnostic.

Read the subsections in clusters rather than as independent claims. Oracle Results frames baseline behavior under known evidence; the noisy component gate and same-stream comparison carry Claim 1. The lookup diagnostic and released-artifact PFLC replay carry Claim 2, while the LongMemEval v1 readout-cardinality probe carries Claim 3. The all-hit floor is a scoring-property bridge that bounds the follow-up, and the pending multi-evidence follow-up carries Claim 4.

4.1 Oracle Results

The frozen mechanism-diverse oracle sweep matched all preregistered deltas. CQ and the ADD/UPDATE/NOOP eager-write baseline tied on aggregate answer correctness, false assertion, poison promotion, clean durable displacement, and leakage. CQ's narrower advantage was lower premature promotion, especially on false-corroboration mixed-source rows. The result supports staged promotion as a way to reduce premature durable commitment under oracle inputs; broad answer-quality superiority over the ADD/UPDATE/NOOP eager-write baseline remains outside the supported claim.

The adversarial upstream-noise sweep is similarly mixed. CQ wins retraction, witness conflict, scope narrowing, and pending competition, but loses temporal skew. The post-hoc dated-evidence repair turns base CQ's temporal-skew abstention into a correct answer from fresher dated evidence and preserves the four original wins. The CQ-vs-Reflection win criterion remains unmet because Reflection also lands at the correctness ceiling.

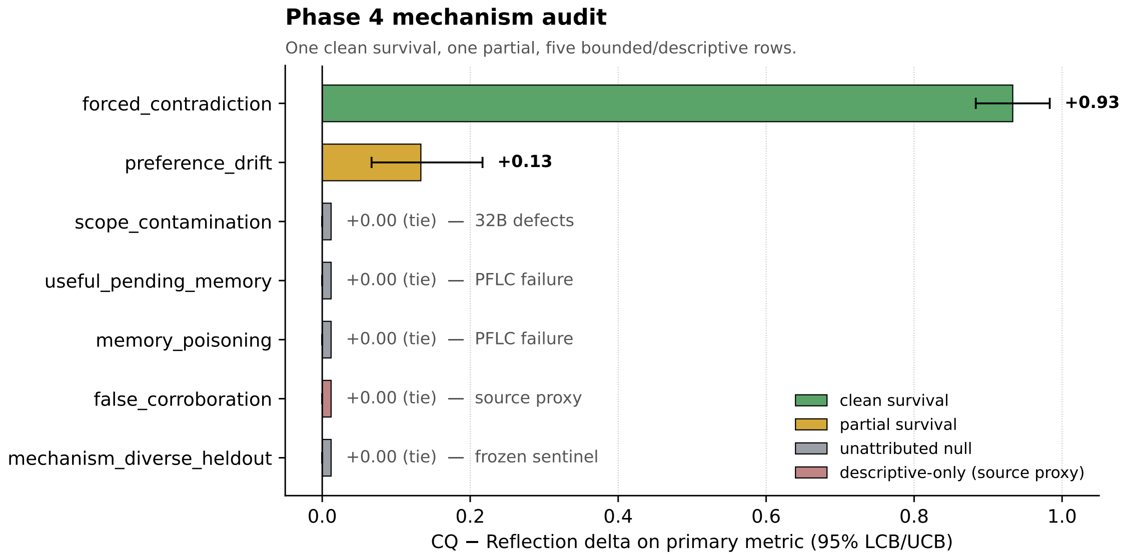

Figure 4: Noisy same-stream policy comparison mechanism audit summary. Only forced contradiction is a clean survival row; preference drift is partial, and other rows remain bounded, diagnostic, or descriptive evidence (component defects, lookup-contract failures, the descriptive-only event-source proxy for false corroboration, or the frozen-sentinel held-out tie).

4.2 Noisy Component Gate and Policy Comparison

The component gate demonstrates why aggregate extractor metrics are insufficient. The locked 7B row passed prompt-regression, determinism, aggregate confidence-bound gates, and aggregate observed gates. It still stayed locked because it had 45 primary scenario errors, eight per-family observed failures, and three frozen-sentinel observed failures. An aggregate-only rule would have authorized a false noisy policy comparison.

The preregistered 32B rows cleared the gate and licensed the noisy same-stream policy comparison. The comparison reaches mixed/partial support: CQ wins against `Reflection EagerWriteLite` on forced contradiction (delta +0.93, lower confidence bound +0.88) and preference drift (delta +0.13, lower confidence bound +0.07), records no countable directional losses, ties the ADD/UPDATE/NOOP eager-write baseline on frozen sentinel primary metrics, and has no replicate contradictions. The mechanism audit narrows this: forced contradiction is the clean mechanism-survival row, preference drift is partial survival, and the remaining rows stay null, descriptive, or bounded by component evidence.

4.3 Lookup Diagnostic and PFLC

The lookup diagnostic adds a second diagnostic layer to several tied rows. On `useful_pending_memory`, `memory_poisoning`, `false_corroboration`, and the frozen sentinel, B-cubed F1 is at or near 1.00 while exact policy-query canonical-id agreement is 0.00. The locked alias-normalized variant also stays at zero on those rows. The diagnostic does not replace the mechanism-audit labels in Figure 4: the `false_corroboration` row remains descriptive-only source-proxy evidence, the mechanism-diverse row remains the frozen sentinel aggregate, and the policy verdict does not change. It shows that the extractor can look cluster-correct while the policy-facing lookup contract fails.

The field-level PFLC work generalizes this observation. Identity-PFLC names the memory-state gap: clustering, retrieval, and answer-accuracy metrics can expose good groups, relevant content, or correct text while failing to certify the policy-facing handle. The formal counterexamples live in Appendix B. The four-anchor survey over LongMemEval, Mem0/LoCoMo, MemoryAgentBench, and MemBench splits on artifact accessibility: most released baseline outputs are artifact-blocked, but a subset of published Agent Memory Benchmark (AMB) LoCoMo runs expose recoverable per-question evidence ids and support a published-output PFLC replay. The split is summarized below; benchmark-targeted synthetic counterexample rows ship alongside.

4.4 PFLC Decomposes Published LoCoMo Memory-Benchmark Outputs

Across the surveyed memory-benchmark artifacts, public outputs divide cleanly into two classes. Most rows—Mem0's `memory-benchmarks` platform, A-MEM/AgenticMemory, SNAP LoCoMo RAG, Backboard, MemoryLake, Engram, and external AMB comparison rows for Mem0, Zep, Letta, LangMem, and MemoryBank—do not commit per-question retrieved context, retrieved memory ids, or LoCoMo evidence ids in their released artifacts, and so admit no published-output PFLC replay. Three AMB LoCoMo runs do: Hindsight (1,540 cases), hybrid-search (1,540), and Cognee (152). Each injects per-question context whose source blocks carry recoverable LoCoMo `dia_id`s, enabling a context-derived PFLC replay against the official LoCoMo `qa[].evidence` field.

Table 4: Published-output PFLC on AMB LoCoMo runs (lookup-relevant denominator).

Run	Scored	Lookup-rel.	PFLC@10	PFLC@20	PFLC@50
AMB `locomo-hindsight`	1,540	1,536	65.8%	83.1%	97.6%
AMB `hybrid-search`	1,540	1,536	64.8%	75.7%	90.5%
AMB `cognee`	152	150	60.0%	73.3%	92.0%

The replayable runs are not lookup-contract collapses. They decompose LoCoMo's aggregate answer accuracy into three components that the aggregate conflates. For example, AMB hybrid-search produces 276 cases where the answer is wrong while every gold evidence `dia_id` is already present in the emitted context by PFLC@50, isolating an answering or judging residual from a retrieval-contract residual. Conversely, 100 cases are correct while all gold evidence is missing at PFLC@50, flagging answer success not fully backed by the LoCoMo evidence contract as scored here. PFLC@10 and PFLC@20 carry the rank-sensitivity signal because Hindsight averages 36,235 context tokens per question.

The operational claim is therefore narrow: once per-question context is published, PFLC decomposes LoCoMo scores into evidence exposure, evidence rank or context saturation, and answer-generation residuals. This is not handle-identity certification; that claim is carried by the internal canonical-id diagnostic. Without per-question artifacts, the same decomposition is impossible, which is why the artifact-blocked subset remains the headline survey gap.

4.5 Controlled Readout-Cardinality Probe on LongMemEval v1

The LongMemEval v1 readout-cardinality probe produces an inspectable negative external result. The run covers 1,935 policy/cell/case rows across `primary_contract`, `path_a_only_denominator`, and `path_b_only_denominator`. Candidate-stream hash invariants pass in all cells. The headline result is shown in Table 5.

The stable sign fires the preregistered-gate condition, but the sign is negative. CQ retrieves at least one gold evidence session in every case and therefore ties eager baselines on `any_hit_at_50`,

Table 5: LongMemEval v1 readout-cardinality probe: CQ-vs-Reflection result.

Cell	all_hit_at_50 delta	any_hit_at_50 delta
primary_contract	-1.0	0.0
path_a_only_denominator	-1.0	0.0
path_b_only_denominator	-1.0	0.0

which asks whether at least one required gold session is returned. It never retrieves both evidence sessions, so it loses all_hit_at_50. The methodology gate passes, but the probe is CQ-negative and isolates a policy-facing pending-readout gap.

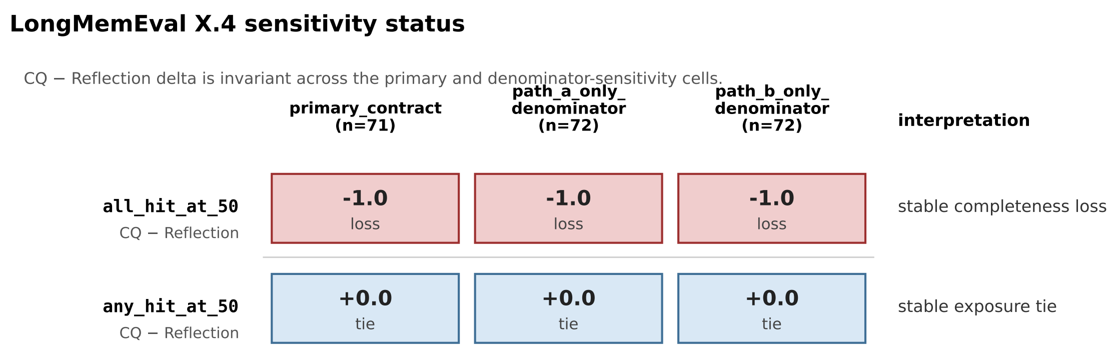

Figure 5: LongMemEval v1 readout-cardinality probe sensitivity status. The CQ-vs-Reflection delta is stable across the primary and denominator-sensitivity cells: -1.0 on all_hit_at_50 and 0.0 on any_hit_at_50.

4.6 All-Hit Metric Floor

The LongMemEval v1 readout-cardinality probe is evidence-only: in the committed rows, the observed unique session-id pool for each scored case contains exactly the two gold evidence sessions. A scorer-only random floor therefore exposes a sharp cardinality property of the metric. A singleton readout has zero chance to satisfy all_hit_at_50; a random two-session readout has a 100% closed-form floor in all three cells, confirmed by five seeded draws per case. This does not weaken the original base-CQ loss, because base CQ returns only one session. It does bound the follow-up interpretation: the plural-pending readout variant succeeds on v1 by exposing plural same-slot evidence, not by demonstrating broad distractor-heavy retrieval quality.

Table 6: Scorer-only random-readout floor for LongMemEval all_hit_at_50.

Cell	Cases	Mean pool	Mean gold	Random r=1	Random r=2	5-seed r=2 band
path_a_only_denominator	72	2.00	2.00	0.0%	100.0%	100.0%–100.0%
path_b_only_denominator	72	2.00	2.00	0.0%	100.0%	100.0%–100.0%
primary_contract	71	2.00	2.00	0.0%	100.0%	100.0%–100.0%

4.7 Pending Multi-Evidence Follow-Up

The follow-up is structured around a paired pre-registration: a CQ-side *repair* hypothesis and a Reflection-side *symmetric falsifier.* The repair hypothesis is that the controlled-probe loss is induced by answer-time readout cardinality, not by CQ-only storage or asymmetric inputs. If that is correct, two things must hold in the same cells. First, the plural-pending readout variant must close the gap on `all_hit_at_50`. Second, the capped Reflection control must reproduce the original base-CQ loss with the same magnitude. The plural-pending readout variant changes the no-durable pending fallback to expose all eligible same-slot pending candidates, with no change to storage, adapter, judge, or headline metric. The capped Reflection control preserves Reflection's winning write path and only caps answer readout to one durable candidate id. The symmetric falsifier is the load-bearing half: it rules out CQ-only storage and asymmetric-input explanations for the loss by reproducing it on a baseline that previously won the metric.

Both predictions hold (Table 7). Plural pending readout moves base CQ from 0/71 to 71/71 in the primary cell. Capping Reflection moves it from 71/71 to 0/71 in the same cell. The pattern is stable across both denominator-sensitivity cells.

Table 7: Pending multi-evidence follow-up on `all_hit_at_50`: the repair and the symmetric falsifier.

Cell	Base CQ	Follow-up CQ	Reflection	Capped Reflection
`primary_contract`	0/71	71/71	71/71	0/71
`path_a_only_denominator`	0/72	72/72	72/72	0/72
`path_b_only_denominator`	0/72	72/72	72/72	0/72

The internal regression gate reports maximum checked overall delta 0.0 across forced contradiction, preference drift, mechanism-diverse held-out, adversarial upstream noise, and evidence-conflict spectrum. The committed metrics are aggregate policy rows rather than per-scenario CQ-Multi-vs-base-CQ pairs, so the regression table reports per-family overall deltas only.

Table 8: CQ-Multi internal regression replay over aggregate policy rows.

Family	Scenarios	Checked metrics	Max $\lvert\Delta\rvert$	Non-zero deltas
Forced contradiction	25	26	0.000000	none
Preference drift	25	26	0.000000	none
Mechanism-diverse held-out	3	26	0.000000	none
Adversarial upstream noise	25	26	0.000000	none
Evidence-conflict spectrum	25	26	0.000000	none

The aggregate maximum checked overall delta is 0.000000. The committed CQ-Multi CSV artifacts contain policy-summary rows, so this table reports per-family overall deltas only; no per-scenario confidence intervals are inferred from these aggregate files.

Focused negative-control tests confirm that below-floor poisoned siblings, contested siblings, wrong-scope siblings, mirrored-source weak siblings, and demoted siblings are not returned by plural pending lookup. With the symmetric falsifier holding, this mechanism-local repair supports answer-time readout cardinality, not a richer CQ-only memory substrate or broader CQ-vs-Reflection superiority.

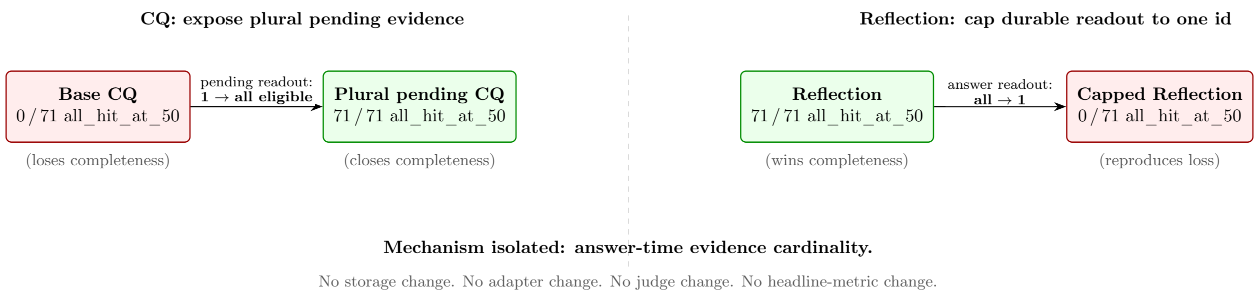

Figure 6: Cardinality-repair follow-up. Changing the CQ pending-readout cardinality from singleton to plural converts a 0/71 loss into a 71/71 win on `all_hit_at_50` in the primary cell. Capping the durable Reflection readout to a single id reproduces the original CQ loss with the same magnitude. The symmetric swap isolates answer-time evidence cardinality as the responsible interface variable; nothing else in the pipeline changes.

5 Discussion

The results map when CQ's mechanisms remain visible. When the noisy stream preserves contradiction edges and the policy-query surface stays aligned, CQ's contestation and demotion behavior survives. When the stream or interface omits canonical slots, source identity, scope handles, or the relevant policy-facing identifier, policy ties become evidence about the benchmark surface rather than about policy equivalence.

The negative and mixed rows carry that evidence. The 7B gate shows that aggregate component metrics can hide concentrated failures, while the 32B mixed-support audit leaves several ties as bounded nulls rather than convergence evidence. The lookup diagnostic shows that clustering-quality metrics can pass perfectly while exact lookup handles fail. The LongMemEval v1 readout-cardinality probe shows that an external adapter can pass methodology gates and still produce a stable CQ-negative result; the pending multi-evidence follow-up then isolates one interface failure mechanism without rewriting the original result.

The main claim is methodological: same-stream memory-policy evaluation makes policy distinctions auditable. It surfaces both positive mechanism survival (forced contradiction under 32B extraction) and cases where an interface hides the distinction, such as exact policy-query canonical-id failure under perfect clustering or evidence-completeness failure under singleton pending readout.

The CQ case study also illustrates why immediate-write baselines must be strong and fair. Reflection sometimes wins for mechanistically uninteresting reasons, such as overwriting into a durable state that happens to include both evidence sessions, but the comparison remains valid because the baseline received the same stream and used the same substrate. The follow-up uses that interpretation directly: the interface change is registered before execution, and the capped-Reflection control rules out CQ-only storage and asymmetric-input explanations by reproducing the original loss on a winning baseline through a single-variable readout change. This isolates answer-time readout cardinality as the mechanism, but does not generalize to a broader metric-fitting defense.

6 Extractor-Gate False-Unlock Check

The extractor gate is an empirical anchor because it shows how aggregate component metrics can falsely authorize downstream memory-policy comparison. It reruns the locked 7B anchor and two

32B primary cells under preregistered model tags, schema profiles, digest checks, and frozen sentinel requirements. The 7B anchor reproduces the locked failure counts: it passes aggregate checks but fails scenario-family and frozen-sentinel checks. Both 32B cells clear the preregistered gate, with zero primary scenario errors, zero primary observed failures, zero aggregate failures, and zero frozen-sentinel observed failures.

This section is intentionally short because the unlock is not itself a memory policy result. Its role is licensing. It shows that the later noisy same-stream policy comparison was not run on an extractor stream that the preregistration would have rejected, while the 7B anchor remains a clean example of an aggregate-passing false unlock that the full gate prevented.

7 Limitations

- **CQ is not broadly superior.** The noisy-mode support is mechanism-local: CQ separates from `ReflectionEagerWriteLite` on forced contradiction and partially on preference drift under the current 32B artifacts, ties the `ADD/UPDATE/NOOP EagerWrite` baseline on the frozen sentinel, and leaves source-independence, scope-contamination, useful-pending, and poisoning-defense rows unsupported. A stronger extractor could produce CQ separations, policy ties, or CQ weaknesses; this revision does not claim an outcome for those rows.
- **LongMemEval v1 is bounded and cardinality-sensitive.** The controlled v1 probe is stable and negative for base CQ on evidence completeness, and the pending multi-evidence follow-up supports a readout-cardinality repair for that v1 loss. The v1 oracle split used here is evidence-only, and the scorer-only random floor shows why returning two evidence sessions is enough in this cell. LongMemEval-S, LongMemEval-V2, distractor-heavy retrieval quality, and natural-language QA success remain future targets.
- **LoCoMo evidence exposure is not handle-identity certification.** The AMB LoCoMo replay demonstrates evidence exposure over gold evidence ids and decomposes aggregate answer accuracy into evidence exposure, rank/context saturation, and answer residuals. It does not certify policy-facing handle identity. Handle-identity failure is demonstrated by the internal canonical-id diagnostic.
- **PFLC is diagnostic, not a leaderboard.** The field-level survey records what released baseline artifacts expose. Most released artifact contracts and standard output surfaces are blocked for PFLC replay, while the AMB LoCoMo subset admits a published-output replay that decomposes aggregate accuracy without resolving the blocked majority.
- **The ADD/UPDATE/NOOP baseline is not production Mem0.** The local baseline is a narrow rule-based comparator over the shared candidate stream and substrate. Production Mem0 exposes richer add, search, update, delete, and platform behavior, so this paper's results do not characterize the production system.
- **Deployment surfaces are not evaluated.** The experiments do not measure real-user weeks-long helpfulness, learned semantic scope inference, production retrieval stacks, API-hosted extractors, deployment latency, deletion, or compliance. A future deletion experiment first needs separable policy-facing query surfaces and registered leakage metrics.
- **The resolution audit remains inconclusive.** The lookup diagnostic leaves the aborted resolution audit and the noisy same-stream policy comparison's mixed-support verdict unchanged. A later archived replay surfaced too few alias-resolution hits to evaluate the lookup-

versus-answer-success cross-tab under its preregistered minimum-count rule; prior PFLC null rows for those families remain unattributed. The archived readout is recorded in Appendix D and Appendix F.

8 Future Work

Closing bounded-null rows under stronger extraction. The noisy same-stream policy comparison leaves four rows—source-independence, scope-contamination, useful-pending, and poisoning—uninformative under the current extractor: the candidate stream collapses the policies to the same measured outcomes. A preregistered extension would rerun these rows with a stronger extractor, either a larger local model or an API-hosted extractor under the same redacted same-stream guarantee. The prediction is that at least scope-contamination and useful-pending separate CQ from Reflection with a 95% lower confidence bound above zero. If not, we can conclude these rows are policy ties under this benchmark, not merely extraction-floor artifacts.

LongMemEval beyond the v1 readout-cardinality probe. The pending multi-evidence repair is scoped to the v1 evidence-only split. The 2026-05-25 LongMemEval-S feasibility gate did not find a local source artifact to pin, so this revision does not run a held-out -S adapter. A future extension must first pin the dataset, inspect the answer/evidence and distractor-session fields, and decide whether filler sessions enter the same candidate stream or only the retrieval surface. If that mechanism-alignment check passes, the extension should keep evidence-id completeness as the primary PFLC target, add `all_hit_at_k` at small k as a precision-sensitive co-primary, and classify the predicted failure mode before unblinding.

External-transfer boundary to conflict resolution. MemoryAgentBench's conflict-resolution tasks are the closest external surface to CQ's contestation and demotion mechanism. A rigorous transfer must satisfy three preconditions before policy execution: an adapter that exposes per-question identifiers sufficient for PFLC scoring; hidden-answer verification analogous to the LongMemEval dual-path protocol; and a preregistered abort criterion if the source benchmark does not commit to a stable evidence id across releases. If any precondition fails, we will preregister the transfer as descriptive-only and will not run policies.

Resolving the resolution audit abort. A fresh archived replay with hash-pinned manifests has replaced the earlier unrecoverable replay path, but it still surfaced too few alias-resolution hits to evaluate the lookup-versus-answer-success cross-tab. The remaining open question is narrower than before: either a preregistered amendment to the alias function, or a stronger extractor cell that surfaces more canonical-id-aliasing candidates, could populate the cross-tab. Until then, the prior PFLC null rows remain unattributed rather than hidden wins.

PFLC as a benchmark output contract. The field-level PFLC survey divides into an artifact-blocked majority and a PFLC-replayable subset (the AMB LoCoMo runs); the blocked subset is what prevents a uniform diagnostic. A concrete next step is a minimal PFLC output schema—per question, the gold evidence id set and the policy-returned id set—offered as a patch to one surveyed benchmark in the blocked subset. The success criterion is adoption by at least one benchmark maintainer. The acceptable failure criterion is principled rejection on the grounds that the schema cannot represent a specific task family. Silent neglect is the failure mode this paper's survey already documents and is not acceptable evidence against the schema.

Deployment realism and policy-facing surfaces. Deletion and compliance are the natural deployment-realism lane for PFLC. The current local policies are not ready for that experiment because they share the same `answer_question` surface and query the substrate directly, rather than exposing separable policy-facing lookup contracts. A future deletion experiment should first define those surfaces and deletion semantics, then preregister leakage metrics before any policy execution.

9 Conclusion

Memory metrics can pass while memory policies fail. The CQ case study demonstrates this concretely at two layers: extractor-gating blocks an aggregate-passing false unlock before policy comparison, and identity-PFLC shows why clustering, retrieval, or answer metrics cannot certify the policy-facing handle needed for governance. The AMB LoCoMo replay adds a published-output evidence-exposure decomposition, while the LongMemEval v1 probe preserves a stable CQ loss that a single-variable repair and a symmetric falsifier isolate as an answer-time readout-cardinality gap. The contribution is the evaluation discipline, not CQ superiority. The discipline is portable to other persistent-agent memory systems, and the CQ artifacts and preregistration locks are released as worked examples of how to apply it.

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A Glossary: Reader-Facing Names and Repository Labels

The paper uses reader-facing descriptive names in body text and introduces the matching repository label once, in parentheses, on first use. This appendix collects the full mapping for readers who encounter a label in the codebase or in committed artifact paths.

Table 9: Glossary: descriptive names and repository labels.

Reader-facing name	Repo label	Meaning
prompt-regression guard	Phase A	Pre-scoring check that the extractor's prompt has not regressed against locked reference outputs.
noisy extractor-quality gate	Phase 3	Component-quality gate that scores local extractor outputs and licenses downstream policy comparison.
noisy same-stream policy comparison	Phase 4	The main noisy-pipeline policy comparison that runs only after the extractor-quality gate passes.
cardinality-repair follow-up	Phase Y	The LongMemEval follow-up that isolates answer-time readout cardinality as the mechanism behind the controlled-probe loss.
LongMemEval v1 readout-cardinality probe	X.4	The controlled-pilot externalization of the same-stream protocol to LongMemEval v1.
preregistered-gate condition fired	Bucket A	A planned gate or thesis condition fired during a preregistered comparison.
mixed/partial support	Bucket B	A completed result that is mixed or partial; rows split between mechanism-survival, partial survival, and bounded nulls.
abort / unsupported claim	Bucket D	A preregistered abort or surprise-lane boundary that prevents the stronger claim.
lookup diagnostic	QR-canon	The canonical-id PFLC instance: a small committed diagnostic table over per-question canonical ids.
resolution audit	CQR	The earlier canonical-id/query-resolution replay audit whose set-membership metric and alias function the lookup diagnostic reuses.
ADD/UPDATE/NOOP eager-write baseline	`Mem0Lite` (repo); `Mem0 WritePolicyLite` (legacy paper label)	Narrow ADD/UPDATE/NOOP write-time baseline; *not* an end-to-end reproduction of the production Mem0 system. Repo identifier kept stable for artifact compatibility; reader-facing body text uses the descriptive baseline name.
oracle-mode dated-evidence repair	`CQDatedContestation` (macro `\textbackslash cqdated`)	Oracle-mode repair variant that prefers dated evidence on temporal-skew rows; used in the adversarial upstream-noise follow-up.

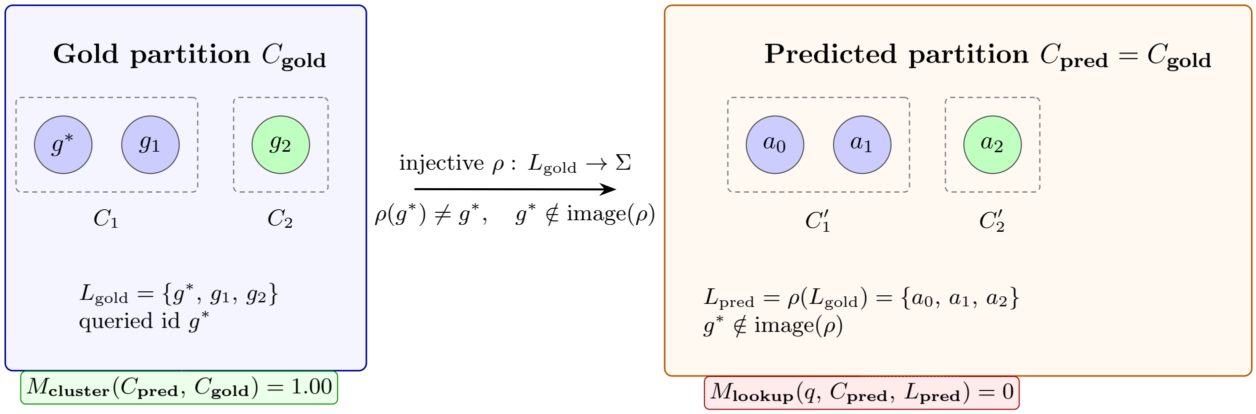

Figure 7: PFLC formal gap (Proposition 1, worked example). The predicted partition is identical to the gold partition, so any cluster-partition metric (M_{cluster}: B-cubed F1, ARI, NMI, V-measure, pairwise cluster F1) scores 1.00. The injective relabeling ρ maps L_{gold} into Σ with image disjoint from $\{g^*\}$, so set-membership PFLC for the queried id g^* is 0. The two outcomes depend on disjoint properties of the same predicted state: partition structure versus label image.

Reader-facing name	Repo label	Meaning
CQ plural-pending readout variant	`CQPendingMultiEvidence` (macro `\textbackslash cqmulti`)	CQ-side follow-up that exposes all eligible same-slot pending candidates at answer time; the repair half of the cardinality-repair follow-up.
symmetric capped-Reflection control	`ReflectionEagerWrite CardinalityCapped` (macro `\textbackslash refcapped`)	Reflection-side control with answer readout capped to one durable candidate id; the symmetric mechanism falsifier in the cardinality-repair follow-up.

The 7B and 32B labels in body text refer to local extractor model cells used by the noisy pipeline (Qwen2.5-7B-Instruct and Qwen2.5-32B-Instruct, both quantized), not memory-policy sizes.

B Formal PFLC Gap

This appendix contains the formal version of the identity-PFLC counterexample used in the body. The main text relies on the idea: content or cluster quality can look correct while the policy-facing handle needed for governance is missing. The figure records the corresponding relabeling construction.

C Failure Taxonomy

The noisy extractor-quality gate taxonomy groups component failures into four recurring modes: required-field omission, ID-namespace confusion, durable-claim drift into temporary/session framing, and contradiction-edge misses. These categories are used to explain why the 7B aggregate-passing run stayed locked and why some 32B rows remain unsafe to interpret as policy convergence.

Representative inspection traces are:

- `data/results/audit_trace_forced_contradiction_default.html`

- data/results/audit_trace_scope_contamination_default.html
- data/results/audit_trace_useful_pending_memory_default.html

The traces are qualitative inspection aids. Numeric claims come from metrics, manifests, component summaries, and compact audit evidence.

D Reproducibility

The full source repository, frozen at the tag matching this paper, is at https://github.com/contromo/consolidation-queue/tree/paper-v3. The paper/repro/README.md entry in that tree is the reviewer entry point and lists local verification commands.

The repository records provenance at several layers: preregistration lock hashes, adapter pins, model tags and digests, prompt/schema profiles, candidate-stream hashes, clean-worktree checks, Python version, commands, and artifact hashes. Small headline verification artifacts are committed. Large per-scenario run payloads are regeneratable-only unless a preregistration requires byte-stable replay.

The paper was prepared on branch paper_revisions, based on:

longmemeval-externalization-gate

The release plan allowed either merging the LongMemEval follow-up branch into main or pinning the paper package to the exact branch and commit. The paper package uses the tag-pinning path so the arXiv source archive, rendered PDF, and reviewer instructions resolve to the same revision. The reproducibility README in paper/repro/README.md records that decision, the final pinned tag, and the commands for local verification.

The resolution audit replay aborts are reproducibility evidence. The path-normalized repair preserved stable checks and still refused to emit a verdict when locked run JSON payloads could not be reproduced. The paper treats that as support for strict replay discipline. The subsequent publication-hardening Step 3 fresh archived replay (docs/publication_hardening_step3_results.md) continues that lineage: it executed under preregistered preconditions, archived per-family source-run hashes alongside its own outputs in data/runs/publication_hardening_step3_manifest.json, and emitted Bucket B_inconclusive_min_n rather than rescuing a verdict.

The revision package also includes two deterministic report-generation scripts for the CQ-Multi aggregate regression table and the LongMemEval all-hit random floor. Both consume committed artifacts only and do not execute memory policies. The LongMemEval-S and PFLC-surface feasibility gates are recorded as no-go/design-note artifacts rather than silently authorizing new policy runs.

E Preregistration Locks

The main preregistration and lock documents are:

- docs/preregistered_memory_governance_evaluation_template.md
- docs/preregistration.md
- docs/local_unlock_probe_preregistration.md
- docs/noisy_policy_comparison_preregistration.md

- docs/adversarial_upstream_noise_preregistration.md
- docs/adversarial_upstream_noise_dated_followup_preregistration.md
- docs/abstention_quality_preregistration.md
- docs/canonical_id_resolution_audit_preregistration.md
- docs/canonical_id_resolution_audit_repair_preregistration.md
- docs/qr_canon_audit_registration.md
- docs/policy_facing_lookup_contract_registration.md
- docs/fair_stream_externalization_preregistration.md
- docs/cq_pending_multi_evidence_preregistration.md
- docs/publication_hardening_preregistration.md

The pending multi-evidence follow-up lock SHA is:

```
84b3c44148494c1b6f72e8201a8da886f8e1f48e49ce00197f535c7e08638840
```

The publication-hardening lock SHA is:

```
05671290ae8ebcc665c915521128c18c48b13be075ace55472b5931874672590
```

The canonical-id PFLC alias function remains the byte-locked resolution-audit alias function; the PFLC workstream does not generalize that alias rule to retrieval-id, slot-id, fact-id, or answer-handle instances.

F Artifact Index

The most important artifact paths for this paper are:

- docs/predictions_vs_results.md
- docs/adversarial_upstream_noise_results.md
- docs/adversarial_upstream_noise_dated_followup_results.md
- docs/abstention_quality_results.md
- data/results/component_gate_decision_general_v1_summary.json
- data/results/component_gate_decision_qwen2_5_7b-instruct-q4_K_M_default_summary.json
- data/results/component_gate_decision_qwen2_5_32b-instruct-q4_K_M_default_summary.json
- data/results/component_gate_decision_qwen2_5_32b-instruct-q4_K_M_scenario_conditioned_summary.json
- docs/noisy_policy_comparison_results.md
- data/results/noisy_policy_comparison_summary.json
- docs/noisy_policy_mechanism_audit.md
- data/results/noisy_policy_mechanism_audit_evidence.json

- docs/canonical_id_resolution_audit_results.md
- data/results/canonical_id_resolution_audit_stop_2026-05-15.json
- data/results/canonical_id_resolution_audit_stop_2026-05-16.json
- docs/qr_canon_audit_results.md
- data/results/qr_canon_audit_metrics.csv
- docs/policy_facing_lookup_contract_proposition.md
- docs/qr_canon_field_diagnostic_results.md
- data/results/qr_canon_field_diagnostic_metrics.csv
- docs/locomo_baseline_replay_survey.md
- data/results/locomo_amb_pflc_summary.json
- docs/longmemeval_transfer_results.md
- data/external/longmemeval/transfer_summary.json
- data/external/longmemeval/transfer_per_case_rows.csv
- data/external/longmemeval/transfer_manifest.json
- docs/cq_pending_multi_evidence_results.md
- data/external/longmemeval/transfer_followup_summary.json
- data/external/longmemeval/transfer_followup_per_case_rows.csv
- data/external/longmemeval/transfer_followup_manifest.json
- data/results/cqmulti_per_row_regression_summary.json
- data/external/longmemeval/all_hit_random_floor.json
- docs/longmemeval_s_dataset_pin.json
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- docs/pflc_surface_audit_note.md
- docs/publication_hardening_preregistration.md
- docs/publication_hardening_step3_precondition_skip.md
- docs/publication_hardening_step3_results.md
- data/results/publication_hardening_step3_summary.json
- data/results/publication_hardening_step3_family_rows.csv
- data/runs/publication_hardening_step3_manifest.json

Figures ship in two forms. The five conceptual diagrams (CQ lifecycle, evidence ledger, externalization protocol, PFLC formal gap, cardinality-repair follow-up) are TikZ source files under `paper/figures/` that PDFLaTeX compiles inline at build time. The two quantitative charts (noisy same-stream policy comparison mechanism audit; LongMemEval v1 readout-cardinality cells) are matplotlib output committed as paired PNG/PDF and regenerated from the committed artifacts above by `paper/repro/render_charts.py`. All figure values trace to the documents and result JSONs listed in this index; the AMB LoCoMo PFLC numbers in Section 4.4 come from `data/results/locomo_amb_pflc_summary.json` (SHA recorded in the summary file's manifest).